15. Midterm Review

February 9, 2026 8:51 AM

1. Pareto Distribution

The Pareto distribution with some shape parameter $\alpha > 0$ has CDF and PDF

$$F(x) = 1 - \left(\frac{1}{x+1}\right)^{\alpha} \quad \text{and} \quad \cancel{f(x) = \frac{\alpha}{(x+1)^{\alpha+1}}}$$

for $x \geq 0$. Use the probability integral transform to find a transformation that takes $U \sim \text{Uniform}[0,1]$ to $X \sim \text{Pareto}(\alpha)$.

write $U = F(x)$ and invert.

$$U = 1 - \left(\frac{1}{X+1}\right)^{\alpha}$$

$$\left(\frac{1}{X+1}\right)^{\alpha} = 1 - U$$

$$X = (1-U)^{-1/\alpha} - 1 \rightarrow U^{1/\alpha} - 1$$

Good Answer

because if $U \sim \text{Unif}[0,1]$ then $1 - U \sim \text{Unif}[0,1]$

1. Beyond the Exponential Distribution

Let $Z \sim \text{Exponential}(\lambda)$, which means

$$f_Z(z) = \lambda \exp(-\lambda z) \text{ and } F_Z(z) = 1 - \exp(-\lambda z)$$

for $z \geq 0$.

1. Find the distribution of $X = \log Z$. ← change of variables
2. Use the probability integral transform to find a transformation from $U \sim \text{Uniform}[0,1]$ to X. ← same as above

In class, we considered $Z = -\log U$

(1)
$$F_X(t) = P(X \leq t)$$
$$= P(\log Z \leq t)$$
$$= P(Z \leq e^t)$$
$$= F_Z(e^t) = 1 - \exp(-\lambda e^t)$$

(2) Invert $F_X(X) = U$

$$U = 1 - e^{-\lambda e^X}$$
$$1 - U = e^{-\lambda e^X}$$
$$\log(1-U) = -\lambda e^X$$
$$-\frac{1}{\lambda}\log(1-U) = e^X \Rightarrow X = \log\left(-\frac{1}{\lambda}\log(\overbrace{1-U}^{U})\right)$$

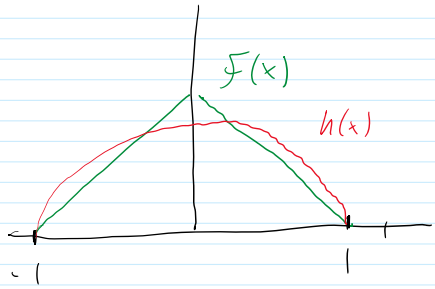
2. Triangular Distribution

The triangular distribution has CDF and PDF

$$F(x) = \begin{cases} (1+x)^2/2, & x \in [-1,0] \\ 1-(1-x)^2/2, & x \in [0,1] \end{cases} \quad \text{and} \quad f(x) = \begin{cases} 1+x, & x \in [-1,0] \\ 1-x, & x \in [0,1] \end{cases}$$

for $x \in [-1,1]$. The acceptance-rejection methods requires us to write $f(x) = cg(x)h(x)$. If we use acceptance-rejection with the the spherical law, $h(x) = (3/4)(1-x^2)$, find c and $g(x)$ and the probability of an acceptance.

$= 1/c$



Step 1: $c = \max_{x \in [-1,1]} \frac{f(x)}{h(x)}$

$(1-x^2) = (1-x)(1+x)$

$$= \max \begin{cases} (1+x) / \frac{3}{4}(1-x^2) & x \in [-1,0] \\ (1-x) / \frac{3}{4}(1-x^2) & x \in [0,1] \end{cases}$$

$$= \frac{4}{3} \left[\max \begin{cases} \frac{1}{1-x}, & x \in [-1,0] \\ \frac{1}{1+x}, & x \in [0,1] \end{cases} \right] \begin{matrix} \leftarrow \text{max at } x=0 \\ \leftarrow \text{max at } x=0 \end{matrix}$$

$= 4/3$

Step 2:

$\text{Prob}(\text{Accept}) = 1/c = 3/4$

$$g(x) = \frac{1}{c} \frac{f}{h} = \begin{cases} \frac{1}{1-x}, & x \in [-1,0] \\ \frac{1}{1+x}, & x \in [0,1] \end{cases}$$

2. Chi Squared Acceptance Rejection

Let $X \sim \chi^2(4)$, which has PDF

$$f(x) = \frac{1}{4} x e^{-x/2}.$$

Use acceptance-rejection to simulate $\chi^2(4)$ random variables from the Exponential $(1/4)$ distribution, which has PDF

$$h(x) = \frac{1}{4} e^{-x/4}.$$

That is, for $f(x) = cg(x)h(x)$, find c, $g(x)$, and the probability of acceptance.

$$c = \sup_{x \geq 0} \frac{f(x)}{h(x)} = \sup_{x \geq 0} \left\{ \frac{\cancel{\frac{1}{4}} x e^{-x/2}}{\cancel{\frac{1}{4}} e^{-x/4}} \right\}$$

$$= \sup_{x \geq 0} \left\{ x e^{-x/4} \right\} \; *$$

$$\frac{d}{dx} x e^{-x/4} = e^{-x/4} - \frac{1}{4} x e^{-x/4} \overset{\text{set}}{=} 0$$

$$e^{-x/4}\left(1 - \frac{1}{4}x\right)$$

$$\frac{d}{dx} \qquad e^{-x/4}\Big(\underbrace{1 - \tfrac{1}{4}x}_{x=4}\Big)$$

* $c = 4e^{-1} = \frac{4}{e} > 1$, $\text{Prob}(\text{Accept}) = \boxed{\frac{e}{4}} \approx \frac{2.71}{4}$

$$g(x) = \frac{1}{c}\,\frac{f}{h} = \frac{1}{4/e}\,x e^{-x/4} = \frac{1}{4}\,x e^{1-x/4}$$